

Whole exome sequencing and RNA sequencing analyses of PDAC samples

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Method Article

Keywords:

Posted Date: January 19th, 2023

DOI: <https://doi.org/10.21203/rs.3.pex-2062/v1>

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Abstract

The associated publication reports proteogenomic analysis of human pancreatic ductal adenocarcinoma (PDAC), where we provided significantly mutated genes (SMGs)/biomarkers, cellular pathways, and cell types as potential therapeutic targets to improve stratification of patients with PDAC. This protocol describes the detailed methods for whole exome sequencing (WES) and RNA sequencing analyses of PDAC samples, including tissue processing for proteogenomic analysis, DNA and RNA extraction, experimental procedures for WES and RNA sequencing, and WES and RNA sequencing data analyses.

Introduction

Reagents

Equipment

Procedure

Tissue processing for proteogenomic analysis

To remove the discrepancies in tissues acquired for genomic, proteomic, and pathological analyses, we adopted a new tissue processing strategy. Briefly, the tumour tissues were cryo-pulverised to obtain a fine tissue powder. The tissue powder was then divided into aliquots for DNA and RNA (ca. 3 mg) isolation and protein extraction (15 mg). This strategy allowed us to produce all proteogenomic data (e.g. WES, mRNAseq, global proteome, and phosphoproteome) and acquire tumour cellularity data from an equivalent tissue sample.

DNA and RNA extraction

DNA and RNA were extracted according to the Allspin (GeneAll) total DNA/RNA purification protocol. Briefly, the tissue powder was immersed in RNAlater reagent (R0901, Sigma-Aldrich) immediately after homogenization. Buffer CTL (+ β -mercaptoethanol; 350 μ L) and tungsten carbide beads (3 mm) were added to the sample tubes; the tissue powder was further lysed by TissueLyzer at 25 Hz for 3 min. The lysate was incubated for 10 min at room temperature, and the lysate in the spin column was centrifuged at 10,000 g for 3 min to separate the DNA and RNA. The pass-through was used for RNA extraction, and the remaining lysate in the mini spin column was used for DNA extraction. The concentration and integrity were measured using a bioanalyzer (Agilent) before library construction.

WES

The SureSelect Human All Exon v5 probe set was used for exome capture according to the Agilent SureSelect Target Enrichment protocol (version B.3, June 2015). The exome was captured by hybridisation of >100 ng genomic DNA onto the capture probes for 24 h on a thermal cycler. Quantification and quality control were conducted using the TapeStation DNA ScreenTape D1000 (Agilent) and quantitative PCR (qPCR) after library construction. Libraries were sequenced with paired end 101 bp reads using the Illumina HiSeq 2500 platform. To call somatic variants in low tumour cellularity samples that included germline genotypes in high proportions, WES depths of blood and tumour samples were produced with at least 100× and 300×, respectively. Reads of FASTQ files were mapped to the GRCh38 genome using BWA (version 0.7.17) with the MEM algorithm¹. Using Samtools (version 1.4.1)², SAM files were further processed to generate, sort, and index BAM files. After alignment, duplicate reads were removed using Picard (version 2.18.7)³. IndelRealigner and BaseRecalibrator modules of the Genome Analysis Tool Kit (GATK; version 3.8)⁴ were used for pre-processing before variant calling. Germline variants in blood samples were identified by HaplotypeCaller, and somatic variants were called using Mutect2 (version 3.8)⁵ from the tumours, when compared with matched blood samples, according to the GATK best practice.

Genomic data analysis

SMGs were identified from the VCF files using MutSigCV (version 1.4)⁶. The exonic variants were annotated using ANNOVAR (version 2018Apr16)⁷, and annovar tables were converted to MAF files and visualized using maftools⁸ R package. Coverage and covariate files provided by MutSigCV were used to calculate q values. Mutated genes with a q value < 0.05 were defined as the SMGs. When determining SMGs, biallelic somatic single nucleotide variants (SNVs) were not considered. For CNV analysis, CNVkit (version 0.9.6)⁹ was applied to the WES data to obtain segmented copy data. Segmented data that did not correspond with the contigs of 22 major autosomes and sex chromosomes were filtered out. The filtered data were analysed using GISTIC2.0 (version 2.0.22)¹⁰ to identify copy number-altered genomic regions and genes. These variants were categorised into missense and nonsense mutations, in-frame insertions, in-frame deletions, frameshift insertions, and frameshift deletions according to the snpEff (version 4.3t)¹¹ annotation. Exome coverage was defined as shared genomic regions between capture probe regions and callable regions using the GATK CallableLoic tool. Tumour mutation burden was defined as the number of protein sequence-altering variants divided by the exome coverage.

RNA sequencing

To obtain high-quality data and overcome RNA degradation problems in the PDAC tissues, exome capture RNA sequencing was performed. After quantifying total RNA using Quant-IT RiboGreen, the samples were run on the TapeStation RNA ScreenTape to calculate DV200, a percentage of RNA fragments of over 200

nucleotides. The mRNA encoding exome was isolated from >100 ng of total RNA using the Illumina TruSeq RNA exome according to the manufacturer's instructions. The strand-specific cDNA library was constructed with adaptor ligation and then sequenced with paired-end 101 bp reads using the Illumina HiSeq 2500 platform. Adaptors and reads of low quality or short length were trimmed using Trimmomatic (version 0.36)¹². The trimmed reads were aligned to the GRCh38 genome using STAR (version 2.5.3a)¹³. The resulting mapped reads were used to quantify the expression levels for genes according to GENCODE v27 GTF annotation using RSEM (version 1.3.0)¹⁴.

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